

Sanskar Modi

sanskarmodi.dev@gmail.com | LinkedIn | GitHub | Portfolio

Professional Summary

AI engineer specializing in production agentic systems, LLM applications, hybrid RAG pipelines, and end-to-end ML pipelines. Contributed 2 merged algorithms to sktime (50k-user open-source library). Built and deployed production AI systems across agent orchestration, medical imaging, reinforcement learning, and retrieval. Seeking AI roles where production quality and technical correctness matter.

Technical Skills

LLM & AI Agents: Google ADK, LangChain, RAG Systems, Tool Orchestration, LiteLLM, ReAct Pattern

ML/DL: PyTorch, Scikit-learn, Deep Learning, Computer Vision, Reinforcement Learning

Production & Cloud: FastAPI, Docker, Google Cloud Run, Azure Portal, CI/CD

MLOps & Tools: MLflow, DVC, Git, Streamlit, HuggingFace, Gradio

Languages: Python, Java

Professional Experience

Open Source Contributor — sktime 2024-2025

Python's leading time series ML library — 50,000+ users, 3,000+ GitHub stars

- Merged **2 production-ready forecasting algorithms** (bias-adjusted Box-Cox + DA-RNN forecaster) into main branch (PR #7909); code reviewed through full CI pipeline with type-hint compliance

AI Engineering Intern — LearnNex (Powered by Wipro) 2025-2025

- Built **production RAG system** with hybrid retrieval (BM25 + dense embeddings) achieving high-precision citation tracking; deployed as inference API via FastAPI
- Developed and fine-tuned **Stable Diffusion pipeline** with custom image generation; delivered both systems end-to-end from model to production

Key Projects

Leave Policy AI Agent — Production Agentic System Live | Code

- Architected production AI agent using **Google ADK + ReAct pattern** on Google Cloud Run; **two-path hybrid architecture**: sub-100ms deterministic fast-path handles **~70% of queries at zero LLM cost**, agentic path handles complex multi-turn reasoning
- Built security layer: **PII detection/redaction**, prompt sanitization, adversarial input filtering; circuit breaker with automatic fallback; **80%+ test coverage** across unit, integration, and security attack tests

AtlasRAG — Hybrid RAG with Knowledge Graph Live | Code

- Hybrid retrieval: dense embeddings + BM25 + **concept co-occurrence graph expansion**; achieved **Recall@5 = 1.00** on multi-section academic document QA evaluation
- FastAPI backend with semantic chunking, citation-aware generation, query rewriting, and Qdrant vector search; deployed on Hugging Face Spaces + Vercel

Brain Tumor Segmentation — Medical CV App | Code

- Multi-modal MRI segmentation using **Swin UNETR** on **40GB BraTS 2024 GLI dataset**; **99.94% accuracy** (avg loss 0.0121) with reproducible DVC pipeline across train/val/test splits

MarketMind — RL Trading System App | Code

- PPO agent achieving **203.8% total return vs 124.1% buy-and-hold baseline** on Bitcoin backtesting; custom reward shaping with transaction costs and turnover penalty
- MLflow experiment tracking + DVC pipeline versioning; **automated weekly retraining** with latest market data via single-command pipeline execution

Education

B.Tech in Computer Science & Engineering

Bennett University, India

Expected Graduation: May 2027